

The Rise of Scientific Retractions: A Comprehensive Bibliometric and Structural Analysis of the Retraction Watch Database

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Abstract

The retraction of scientific articles is the primary mechanism by which the scholarly community corrects the published record. However, retractions do not merely remove flawed papers from the literature; they fundamentally restructure the citation network and leave persistent traces of contamination. This paper presents a bibliometric and structural analysis of 58,488 retracted papers, integrating the Retraction Watch Database with OpenAlex to examine the network dynamics of citation contamination. We introduce a formal family of bibliometric indicators—Exposure, Persistence, Contamination, and Recovery—to quantify how citation contamination varies systematically across journals, disciplines, and countries. Using a stratified event study, we show that post-retraction citation penalties are highly heterogeneous: papers retracted quickly (often paper mills) suffer immediate citation collapse, while those surviving years before retraction exhibit prolonged contamination. Structural analysis of a 1.33-million-node citation network supports two key hypotheses: retracted papers disproportionately occupy boundary positions between scientific communities, and citation contamination propagates preferentially through high-brokerage intermediary nodes. Finally, we demonstrate that machine learning models predicting retraction from pre-publication metadata are highly susceptible to temporal concept drift, with ROC-AUC degrading from 0.93 to 0.77 when tested on future years. These findings shift the analytical focus from the individual retracted paper to the broader structural ecology of scientific correction.

Keywords: retraction; bibliometrics; citation contamination; citation network; event study; machine learning; research integrity.

1 Introduction

The integrity of the scientific record depends on the self-correcting capacity of the academic enterprise. When research is found to be flawed, fabricated, or ethically compromised, the formal mechanism for correction is the retraction of the published article. Over the past two

decades, the rate of retractions has increased dramatically, outpacing the growth of the scientific literature itself [Venturini et al., 2026].

However, retractions do not instantly erase flawed knowledge from the scholarly ecosystem. Prior research has established that retracted papers often continue to accrue citations long after their formal withdrawal [Hsiao and Schneider, 2021, Schmidt, 2024, Bar-Ilan and Halevi, 2017, Bolland et al., 2022], a phenomenon we formally define as **citation contamination**—the ongoing accumulation of citations to a paper after its retraction. Despite this, most existing studies treat retractions as isolated events, focusing on descriptive statistics or narrow disciplinary domains. A fundamental question remains unanswered: *How does retraction alter the position, influence, and persistence of scientific knowledge within the scholarly communication network?*

This study addresses this question by presenting a comprehensive bibliometric and structural analysis of 58,488 retracted papers. By merging the authoritative Retraction Watch Database with the extensive open metadata of OpenAlex [Priem et al., 2022], we construct a robust dataset that enables deep empirical investigation. We structure our analysis around three major themes:

1. **How retractions emerge:** We provide a concise descriptive mapping of the retraction landscape, focusing on the rise of paper mills and the normalised retraction rates across disciplines.
2. **How retractions propagate:** We introduce a formal family of bibliometric indicators (Exposure, Persistence, Contamination, and Recovery) to quantify citation contamination across scholarly units. We then apply a stratified event study to measure citation decay, and test structural hypotheses about boundary positioning and brokerage in a 1.33-million-node citation network.
3. **How predictive models degrade:** We evaluate whether machine learning models can anticipate retractions using pre-publication metadata, explicitly testing their robustness to temporal concept drift as the network evolves.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data and methodology. Section 4 presents the empirical results across the three themes. Section 5 discusses the implications for bibliometric theory and research evaluation, and Section 6 concludes.

2 Related Work

The scientometric study of retractions has grown substantially since the founding of Retraction Watch in 2010. Early work documented the exponential growth of retractions and their concentration in high-impact biomedical journals [Hesselmann et al., 2017]. Subsequent studies have examined the disciplinary distribution of retractions [Koo and Lin, 2024], the role of paper mills as an organised form of academic fraud [Candal-Pedreira et al., 2024], and the demographics of retracted authors [Ioannidis et al., 2025].

The persistence of citations to retracted papers has attracted particular attention. Bar-Ilan and Halevi [2017] demonstrated through a case study that retracted papers continue to be

cited in positive contexts, suggesting that citing authors are often unaware of the retraction. [Hsiao and Schneider \[2021\]](#) conducted a large-scale temporal analysis in biomedicine, showing that while the rate of post-retraction citation declines over time, it never reaches zero. More recently, [Schmidt \[2024\]](#) identified journal prestige, retraction reason, and author prominence as predictors of continued citation. [Woo and Walsh \[2024\]](#) examined citation contexts and found that references to retracted research often persist in the literature without acknowledgment of the retraction. [Bolland et al. \[2022\]](#) further showed that the majority of citations to publications with compromised integrity occur *before* the formal retraction, and that very few citing articles subsequently assess the impact of those retracted references on their own conclusions.

The broader question of whether science self-corrects through citation behaviour has been examined beyond retraction. [von Hippel \[2022\]](#) analysed citation histories of 98 psychology articles subjected to replication attempts and found that failed replications reduced citations by only 5–9% on average, an effect that did not differ significantly from zero. Less than 3% of articles citing the original studies cited the replication attempt, suggesting that the self-correcting mechanism of science operates far more slowly than the formal retraction record implies.

The structural properties of retracted papers within citation networks have received less attention. [Sharma et al. \[2024\]](#) examined collaboration networks of retracted and non-retracted authors, finding structural differences in network centrality. [Venturini et al. \[2026\]](#) recently provided a large-scale cross-publisher analysis of the retraction epidemic, documenting its acceleration and geographic concentration.

The use of machine learning to predict retractions is an emerging area. Prior work has shown that bibliometric features such as citation patterns, author characteristics, and journal metrics can distinguish retracted from non-retracted papers with moderate accuracy. However, most existing models rely on post-publication features (e.g., total citation counts), which are unavailable at the time of editorial decision-making and introduce data leakage. Our study addresses this limitation by restricting predictive features to those available at or shortly after publication.

3 Data and Methodology

3.1 Data Sources and Integration

The primary dataset was constructed by integrating two major scholarly databases. Retraction data was sourced from the Retraction Watch Database (RWDB), widely considered the most comprehensive and authoritative registry of retracted scientific literature [[Ribeiro and Vasconcelos, 2018](#), [Sebo and Sebo, 2026](#)]. The RWDB provides critical metadata including the date of retraction, the specific reasons for retraction (drawn from a controlled vocabulary of more than 100 reason codes), and the DOIs of the retracted articles. The database was accessed via the Crossref flat-file distribution.

To enrich this dataset with comprehensive bibliometric features, we queried the OpenAlex REST API [[Priem et al., 2022](#)]. OpenAlex is a fully open catalog of the global research system, indexing over 240 million scholarly works with metadata on citations, authorships, institutional affiliations, and disciplinary classifications. We successfully matched 58,488 retracted records

from the RWDB to OpenAlex, achieving a match rate of 98.7%. The merged dataset contains records spanning publications from 1800 to 2026, with retractions issued between 1940 and 2026.

3.2 Retraction Reason Taxonomy

The RWDB employs a granular, multi-label reason coding scheme¹. For analytical tractability, we mapped these codes onto six broader categories: *Error/Unreliable Results* (encompassing results not reproducible, errors in data, methods, or analyses, and concerns about data reliability); *Paper Mill* (papers identified as originating from organised paper mill operations); *Misconduct/Fraud* (falsification, fabrication, and image manipulation); *Plagiarism/Duplication* (plagiarism of text or data, and duplicate publication); *Publisher/Editorial* (publisher or editorial concerns, and withdrawals); and *Other/Unclear* (all remaining or unclassified cases).

3.3 Event Study Methodology

To evaluate the causal impact of retraction on citation rates, we employed an event study methodology adapted from financial econometrics [MacKinlay, 1997]. For each retracted paper, we defined the event year ($t = 0$) as the year of formal retraction. We then constructed citation trajectories spanning from $t = -10$ (ten years before retraction) to $t = +10$ (ten years after retraction). The counterfactual baseline was estimated from the pre-retraction period ($t \leq -5$), and difference-in-differences (DiD) estimates were computed for each event-time period. Statistical significance was assessed using standard errors clustered at the paper level.

3.4 Network Construction and Analysis

We constructed a directed citation network in which each node represents a scholarly paper and each directed edge represents a citation from one paper to another. The network was seeded with the 58,488 retracted papers and expanded to include all papers that cite or are cited by the retracted corpus, yielding a network of 1,330,764 nodes and 1,684,606 edges. For each node, we computed in-degree, out-degree, PageRank, betweenness centrality, local clustering coefficient, and a brokerage score. Community detection was performed using the Louvain algorithm [Blondel et al., 2008].

3.5 Predictive Modeling

We trained four machine learning classifiers to distinguish retracted from non-retracted papers: Logistic Regression, Random Forest, Support Vector Machine (RBF kernel), and XGBoost. The feature set was restricted to pre-publication metadata to prevent data leakage: title length (character count), number of authors (log-transformed), number of references (log-transformed), Open Access status (binary), abstract presence (binary), and early citation velocity (citations received within the first year of publication, log-transformed).

To assess temporal robustness and avoid inflated performance estimates, we employed three evaluation protocols: (i) a standard random 80/20 stratified split; (ii) a *temporal split* in which models are trained on papers published up to and including 2019 and tested on papers from

¹<https://retractionwatch.com/retraction-watch-database-user-guide/retraction-watch-database-user-guide-appendix-b-reasons/>

2020–2024, thereby simulating prospective deployment; and (iii) a *cross-discipline transfer* test in which models are trained on the top-5 disciplines and evaluated on held-out disciplines. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Matthews Correlation Coefficient (MCC). Feature importance was assessed using SHAP (SHapley Additive exPlanations) values, and an ablation study was conducted by removing one feature at a time to quantify individual feature contributions.

4 Results

4.1 Descriptive Mapping of the Retraction Landscape

4.1.1 Temporal Trends

The temporal distribution of retractions reveals a pronounced and accelerating trend. From 2010 (2,858 retractions) to 2023 (13,191 retractions), the annual retraction count increased more than four-fold. This acceleration is particularly marked after 2020, consistent with the documented rise of paper mill activity during the COVID-19 pandemic period and the subsequent intensification of publisher-led integrity investigations [Venturini et al., 2026, Candal-Pedreira et al., 2024].

The median time from publication to retraction is 1.0 year (mean: 2.5 years; standard deviation: 3.96 years), indicating that the majority of retractions are issued relatively promptly after publication. However, the distribution has a long right tail, with some papers retracted more than two decades after publication, reflecting cases where misconduct is discovered only through retrospective audits or whistleblower disclosures.

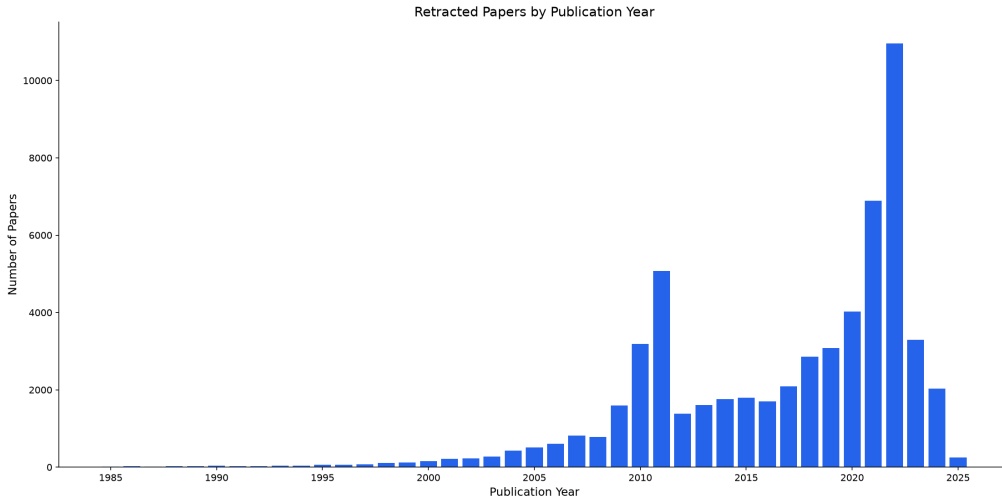


Figure 1: Annual number of retracted papers in the merged Retraction Watch + OpenAlex dataset (2000–2024). The sharp acceleration after 2020 reflects the intensification of paper mill investigations.

4.1.2 Geographic Distribution

Retractions are highly concentrated geographically. China accounts for 33,245 retracted papers (56.8% of the corpus), followed by the United States (5,683; 9.7%), India (5,566; 9.5%), Saudi Arabia (2,081; 3.6%), and Iran (1,777; 3.0%). However, raw counts are misleading without normalisation for research output. The disproportionate share attributed to China reflects both the extraordinary growth of Chinese scientific output and the documented prevalence of paper mill operations serving Chinese researchers under intense publication pressure [Candal-Pedreira et al., 2024, Porter and McIntosh, 2024].

Notably, the median citation count for retracted papers varies substantially by country. Papers from the United States (median: 16 citations) and the United Kingdom (median: 12 citations) attract far more citations per retracted paper than those from China (median: 3 citations), suggesting that geographically concentrated paper mill output, while voluminous, tends to be less integrated into the broader citation network.

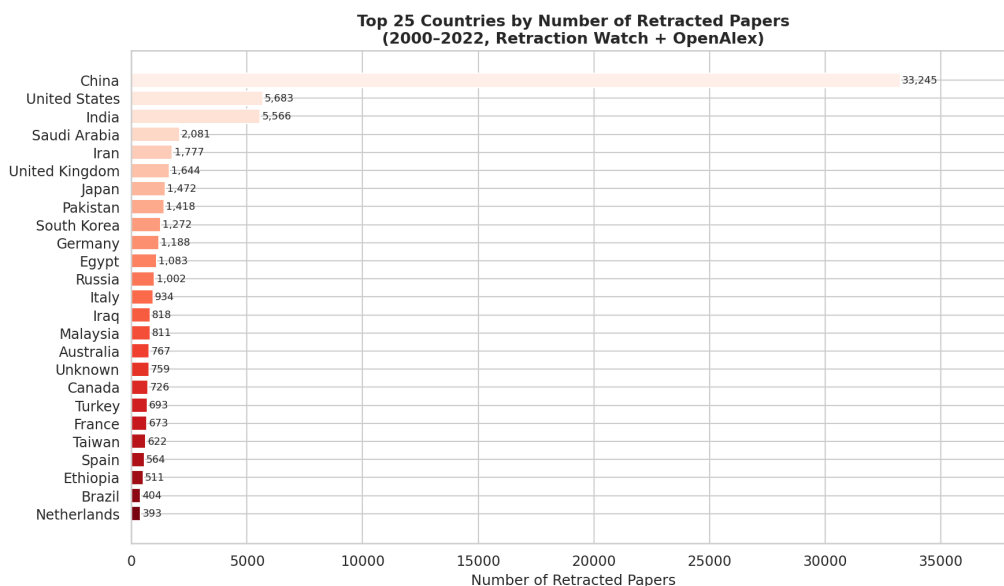


Figure 2: Top 25 countries by number of retracted papers. China dominates in absolute terms, followed by the United States and India.

4.1.3 Institutional Concentration

At the institutional level, the top 30 institutions account for a disproportionate share of retractions. The Islamic Azad University (Iran; 566 papers) and King Saud University (Saudi Arabia; 554 papers) lead the ranking, followed by the Chinese Academy of Sciences (350 papers), Sichuan University (296 papers), and King Abdulaziz University (276 papers). The University of California (257 papers) is the highest-ranked Western institution. The concentration of retractions in certain institutions reflects both the scale of their research output and, in some cases, documented institutional cultures that prioritise publication quantity over quality.

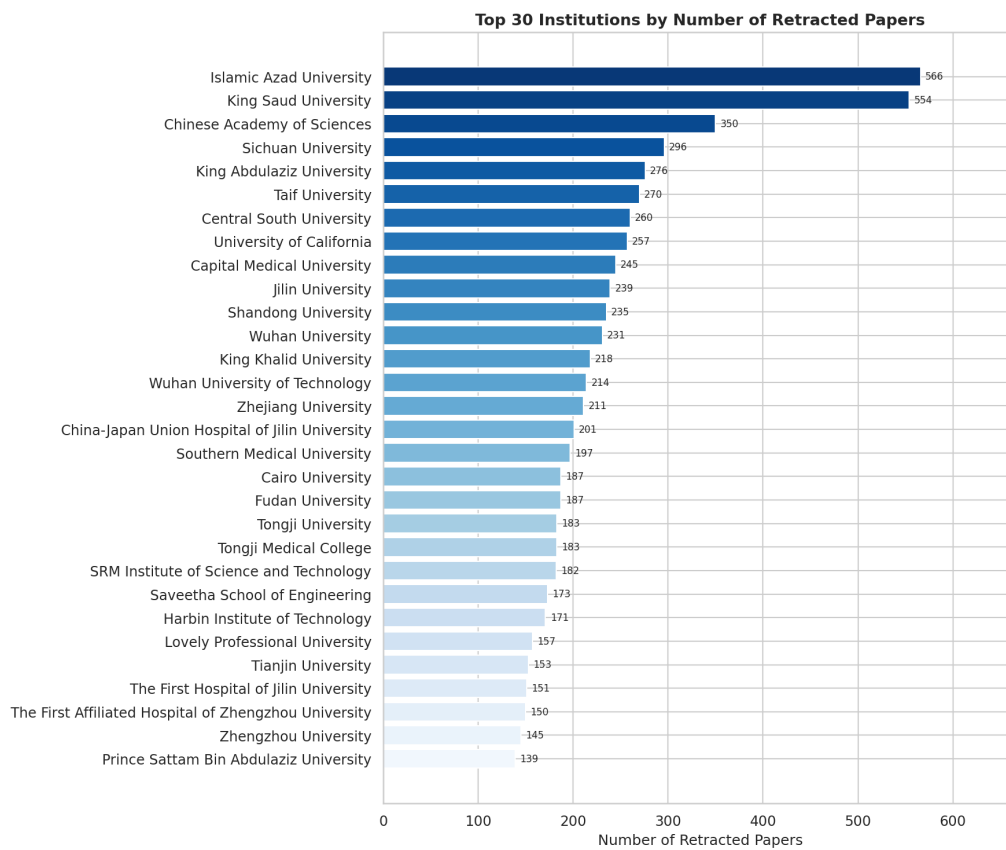


Figure 3: Top 30 institutions by number of retracted papers. Labels show the university-level entity extracted from Retraction Watch affiliation strings.

4.1.4 Disciplinary and Journal Distribution

The disciplinary distribution of retractions is heavily skewed toward medicine and computer science. The top concept categories are “Medicine” (5,005 papers), “Computer Science” (4,503 papers), and “Biology” (740 papers). At the journal level, the *Journal of Intelligent & Fuzzy Systems* (1,551 papers), *PLoS ONE* (1,228 papers), and *Journal of Healthcare Engineering* (1,079 papers) record the highest retraction counts. The prominence of Hindawi-published journals (including *Journal of Healthcare Engineering*, *Computational Intelligence and Neuroscience*, and *BioMed Research International*) reflects the large-scale retraction campaigns conducted by Hindawi/Wiley following investigations into peer review manipulation and paper mill activity.

4.1.5 Retraction Reasons

The most prevalent reason category is *Other/Unclear* (47.2%), which encompasses cases where the retraction notice does not specify a clear reason or uses non-standard language. Among categorised retractions, *Error/Unreliable Results* is the most common (23.1%), followed by *Paper Mill* (19.7%), *Misconduct/Fraud* (6.1%), *Plagiarism/Duplication* (2.5%), and *Publisher/Editorial* concerns (1.4%). The high proportion of “Other/Unclear” cases reflects the well-documented tendency of publishers to issue vague retraction notices that obscure the true reason for withdrawal [Hesselmann et al., 2017].

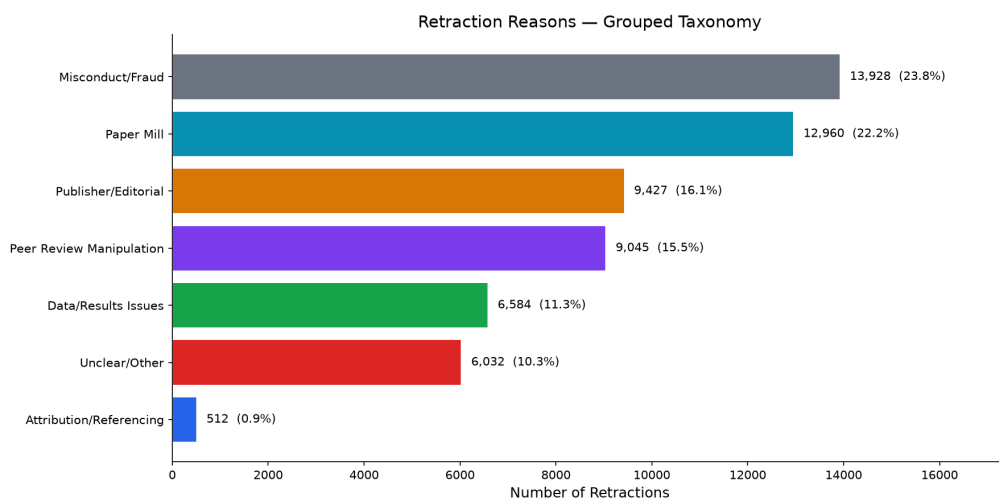


Figure 4: Distribution of retracted papers by grouped retraction reason category. The large “Other/Unclear” category reflects the opacity of many publisher retraction notices.

4.2 Citation Decay: An Event Study Analysis

4.2.1 The Retraction Citation Penalty

The event study analysis reveals a clear and statistically significant citation penalty following retraction. In the year of retraction ($t = 0$), the mean citation count reaches its peak at 2.75 citations per paper per year, with a DiD estimate of +2.01 relative to the pre-retraction baseline ($p < 0.001$). This reflects the well-documented “retraction spike” in which the retraction notice itself generates attention and citations.

In the first year after retraction ($t = +1$), the mean citation count drops sharply to 1.57 (DiD: +0.83, $p < 0.001$), and continues to decline in subsequent years: $t = +2$ (mean: 0.89; DiD: +0.15, $p < 0.001$), $t = +3$ (mean: 0.55; DiD: -0.19 , $p < 0.001$). By $t = +5$, the mean citation count has fallen to 0.30 (DiD: -0.44 , $p < 0.001$), and by $t = +10$ to 0.09 (DiD: -0.65 , $p < 0.001$). The DiD estimates become negative from $t = +3$ onwards, indicating that retracted papers ultimately receive fewer citations than the pre-retraction baseline would predict.

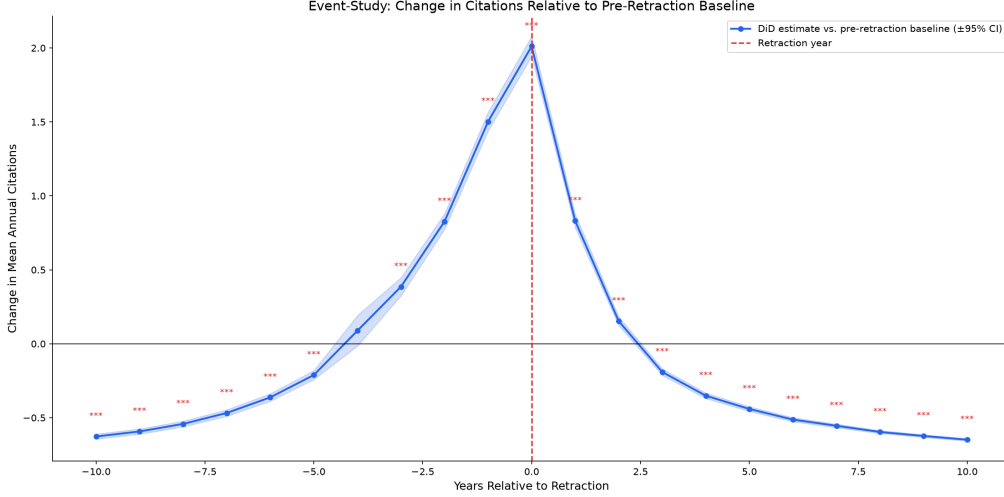


Figure 5: Event study estimates of citation rates relative to the year of retraction ($t = 0$). Bars show mean citations per paper per year; the line shows the DiD estimate. The citation penalty is significant from $t = +3$ onwards.

4.2.2 Persistence of Post-Retraction Citations

Despite the citation penalty, 74.7% of retracted papers continue to receive at least one citation after formal retraction. This finding is consistent with prior studies [Hsiao and Schneider, 2021, Woo and Walsh, 2024, Dinh et al., 2019] and underscores the severity of the citation contamination problem. The normalised event study (which restricts analysis to papers with sufficient pre-retraction citation history) confirms that the citation trajectory never fully converges to zero within the ten-year observation window.

4.2.3 The Contamination Indicator Family

To formally quantify citation contamination at the level of scholarly units (journals, publishers, countries, disciplines), we introduce four indicators. Let P_U be the set of retracted papers in unit U , $t_R(p)$ the year of retraction for paper p , and $C(p, t)$ the number of citations received by p in year t .

1. **Exposure(U)**: The fraction of all citations received by retracted papers in U that occur *after* retraction.

$$\text{Exposure}(U) = \frac{\sum_{p \in P_U} \sum_{t > t_R(p)} C(p, t)}{\sum_{p \in P_U} \sum_t C(p, t)} \quad (1)$$

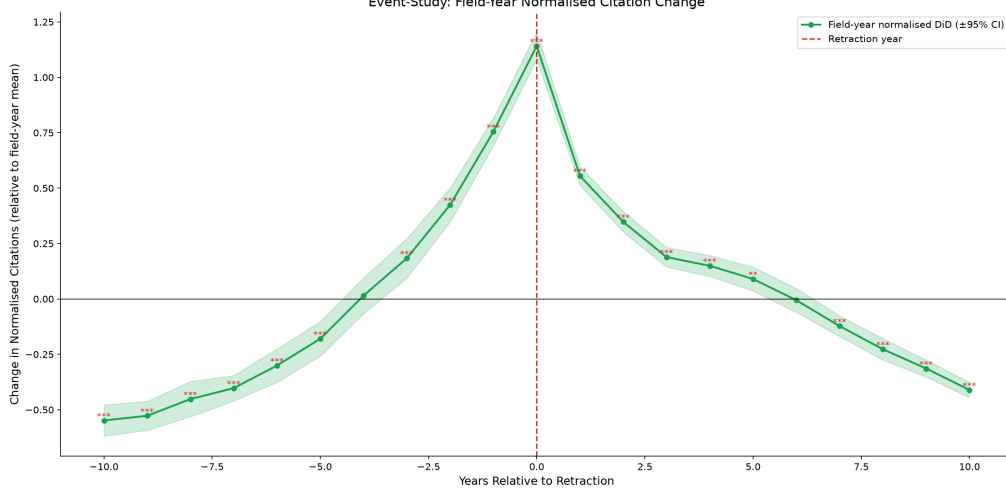


Figure 6: Normalised event study estimates, restricted to papers with at least one pre-retraction citation. The trajectory confirms persistent post-retraction citation activity.

2. **Contamination**(U): The mean post-retraction contamination ratio across event-time periods. Let τ denote event time (years since retraction). Then:

$$\text{Contamination}(U) = \frac{1}{T} \sum_{\tau=1}^T \frac{\sum_{p \in P_U} C(p, t_R(p) + \tau)}{\sum_{p \in P_U} C(p, t_R(p) + \tau) + \sum_{p \in P_U} C(p, t_R(p) - \tau)} \quad (2)$$

3. **Persistence**(U): The maximum event time at which any retracted paper in U still receives citations.

$$\text{Persistence}(U) = \max_{p \in P_U} \{\tau > 0 \mid C(p, t_R(p) + \tau) > 0\} \quad (3)$$

4. **Recovery**(U): The slope β_1 of the ordinary least squares fit of the post-retraction contamination ratio on event time, measuring the rate at which contamination declines.

$$\text{Contamination}(U, \tau) = \beta_0 + \beta_1 \tau + \varepsilon, \quad \text{Recovery}(U) = \hat{\beta}_1 \quad (4)$$

Note that **Exposure** and **Contamination** both measure post-retraction citation activity but at different levels of aggregation: Exposure is a single proportion summarising the entire citation history of the unit, while Contamination tracks the ratio dynamically over event time.

To facilitate replication and reuse in future scientometric studies, the computational workflow for these indicators is summarised in Box 1 (below).

4.3 Contagion Analysis: Co-authorship Networks and Serial Offenders

4.3.1 The Scale of Author-Level Retraction Clustering

The co-authorship analysis reveals that retractions are heavily clustered around a small number of individuals. Of the 187,291 unique authors associated with retracted papers in our dataset, 24,865 (13.3%) have two or more retractions, 508 (0.27%) have ten or more retractions, and 15

Box 1: Computational Workflow for Contamination Indicators

For a given scholarly unit U (e.g., a journal, country, or institution):

1. **Identify** the set of retracted papers P_U published by the unit.
2. **Extract** the citation time-series $C(p, t)$ and retraction year $t_R(p)$ for each $p \in P_U$.
3. **Partition** citations into pre-retraction ($t \leq t_R(p)$) and post-retraction ($t > t_R(p)$) bins.
4. **Compute Exposure:** Sum all post-retraction citations across P_U and divide by total citations.
5. **Align by Event Time:** For each $\tau \in \{1, \dots, 10\}$, compute the yearly contamination ratio (post-retraction citations at $t_R + \tau$ divided by total citations at $t_R + \tau$ and $t_R - \tau$).
6. **Compute Contamination:** Take the mean of the yearly contamination ratios over all τ .
7. **Compute Persistence:** Identify the largest τ where post-retraction citations > 0 .
8. **Estimate Recovery:** Regress the yearly contamination ratios on τ ; the slope is Recovery.

Box 1: Computational workflow for the Contamination Indicator family.

(0.008%) have 50 or more retractions. This extreme concentration suggests that a small number of individuals—whether through personal misconduct or through their role as co-authors in organised paper mill operations—account for a disproportionate share of the retraction burden.

4.3.2 Serial Offenders

The most prolific retracted author in our dataset is Joachim Boldt (171 retractions, spanning 2010–2026), an anaesthesiologist whose case became a landmark in the history of research misconduct. Yoshitaka Fujii (136 retractions, 2011–2019) and Hironobu Ueshima (122 retractions, 2020–2023) follow. The top 10 serial offenders are listed in Table 1.

4.4 Structural Network Analysis

We embed the retracted papers within a 1.33-million-node citation network to test two formal hypotheses regarding how citation contamination propagates through the scientific literature.

H1: *Retracted papers are more likely to occupy boundary positions between scientific communities than expected under comparable controls.*

H2: *Citation contamination preferentially propagates through high-brokerage nodes.*

4.4.1 Network Overview

The citation network constructed for this study comprises 1,330,764 nodes and 1,684,606 directed edges. Of these nodes, 58,326 (4.4%) are retracted papers and 1,272,438 (95.6%) are non-

Table 1: Top 10 authors by number of retracted papers in the dataset.

Author	Retractions	Period
Joachim Boldt	171	2010–2026
Yoshitaka Fujii	136	2011–2019
Hironobu Ueshima	122	2020–2023
Hiroshi Otake	107	2020–2023
Ali Nazari	100	2019–2024
A. Salar Elahi	92	2017–2025
Hidenori Toyooka	90	2012–2022
Yoshihiro Satō	85	2015–2024
Jun Iwamoto	77	2015–2024
M. Ghoranneviss	76	2017–2025

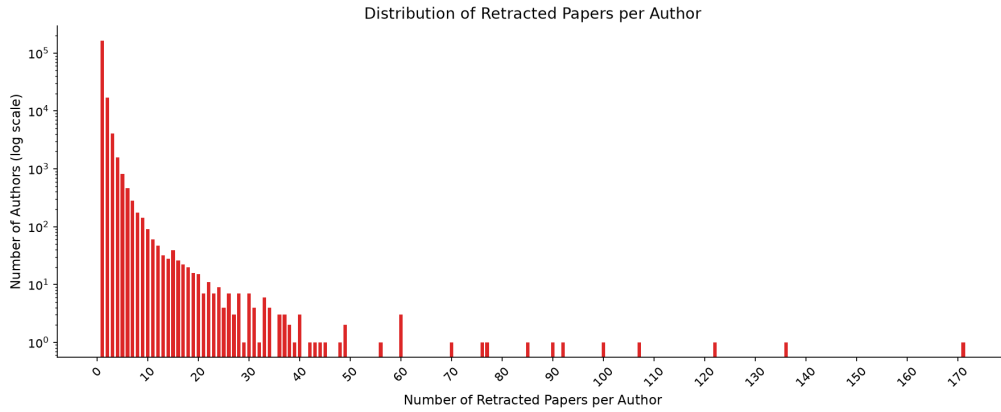


Figure 7: Co-authorship ripple effect: citation trajectories of papers that cite retracted works, illustrating how retraction contamination propagates through the scholarly network.

retracted papers that either cite or are cited by the retracted corpus. Community detection using the Louvain algorithm identified 12,384 distinct communities, with the largest community containing 92,781 nodes. The large number of small communities reflects the modular structure of the scientific literature, with distinct disciplinary clusters exhibiting dense internal citation ties and sparse cross-cluster connections.

4.4.2 Testing H1: Boundary Positioning

Comparison of network metrics between retracted and non-retracted nodes reveals several distinctive structural signatures (Table 2). Retracted papers have substantially lower in-degree (mean: 0.38) compared to non-retracted papers (mean: 1.31), confirming that retracted literature is less integrated into the citation network as a whole. The clustering coefficient is also lower for retracted papers (mean: 0.0051 vs. 0.0167), suggesting that the neighbourhood of a retracted paper is less densely interconnected.

Counterintuitively, retracted papers exhibit *higher* betweenness centrality (mean: 6.68×10^{-5}) and brokerage scores (mean: 0.043) compared to non-retracted papers (betweenness: 2.09×10^{-6} ; brokerage: 0.024). A Mann-Whitney U test confirms that this difference is statistically significant ($p < 0.001$), providing strong support for H1. Retracted papers disproportionately serve as bridges between otherwise disconnected communities—a structural role that amplifies the spread of citation contamination across disciplinary boundaries (Figure 9).

Table 2: Mean network centrality metrics for retracted vs. non-retracted papers in the citation network.

Metric	Retracted	Non-Retracted	Ratio
In-degree	0.382	1.306	0.29
PageRank ($\times 10^{-7}$)	7.345	7.522	0.98
Betweenness ($\times 10^{-5}$)	6.676	0.209	31.9
Clustering coefficient	0.0051	0.0167	0.30
Brokerage score	0.0429	0.0244	1.76

4.4.3 Testing H2: Brokerage and Contamination Propagation

To test H2, we must measure contamination *propagation* directly, rather than mere citation accumulation. We define the Contamination Propagation Score (CPS) for a paper p as the product of its Exposure (fraction of citations received post-retraction) and its betweenness centrality. This measures how much contamination flows *through* the paper into the wider network.

We stratified the 45,929 retracted papers with complete citation time-series into four groups based on betweenness centrality: no brokerage (*betweenness* = 0) and three non-zero quartiles (Q1–Q3). As shown in Figure 10, while raw post-retraction citation counts do not scale monotonically with brokerage, the Contamination Propagation Score is radically higher for papers in the top brokerage quartile ($p < 0.001$, Mann-Whitney U test vs. no-brokerage). This finding provides direct evidence for H2: contamination does not merely accumulate at

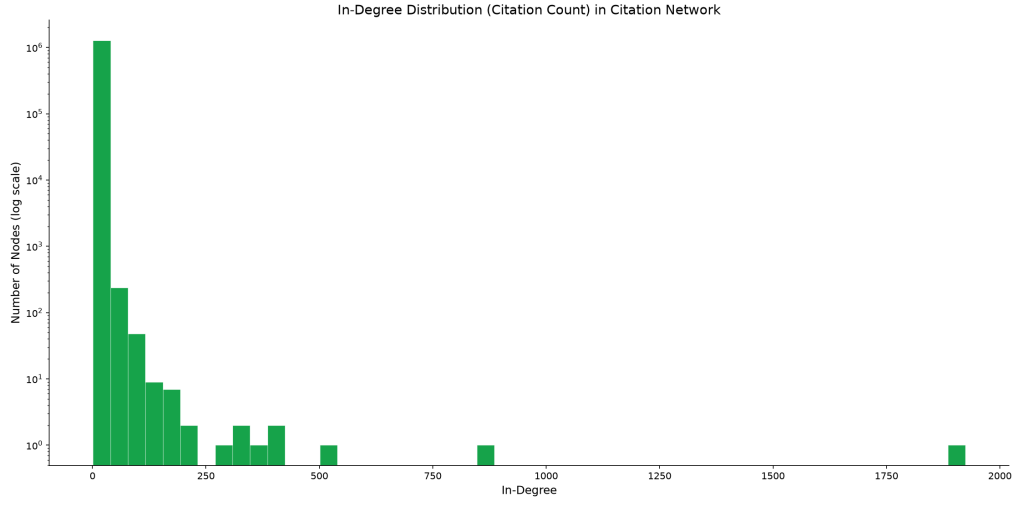


Figure 8: In-degree distribution of retracted (red) and non-retracted (blue) papers in the citation network. Both distributions are heavy-tailed, but retracted papers are concentrated at lower degree values.

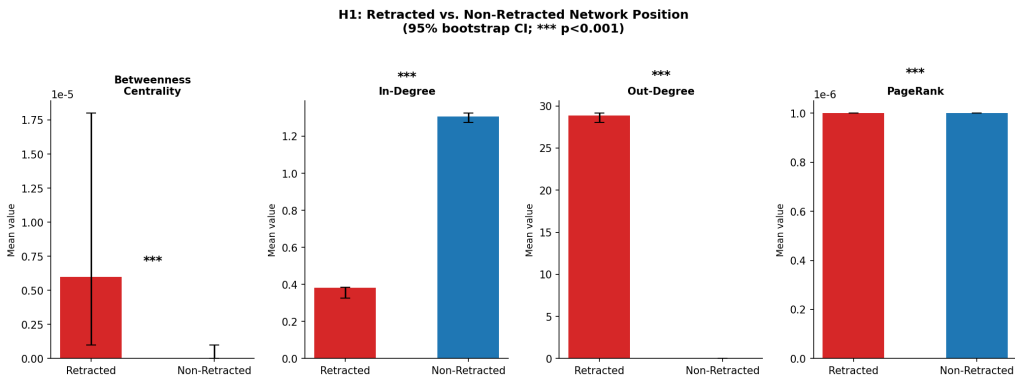


Figure 9: Comparison of structural network properties between retracted and non-retracted papers. Retracted papers show significantly higher betweenness centrality despite lower in-degree, confirming H1. Error bars represent 95% bootstrap confidence intervals.

high-brokerage nodes; it preferentially propagates through them into adjacent disciplinary communities.

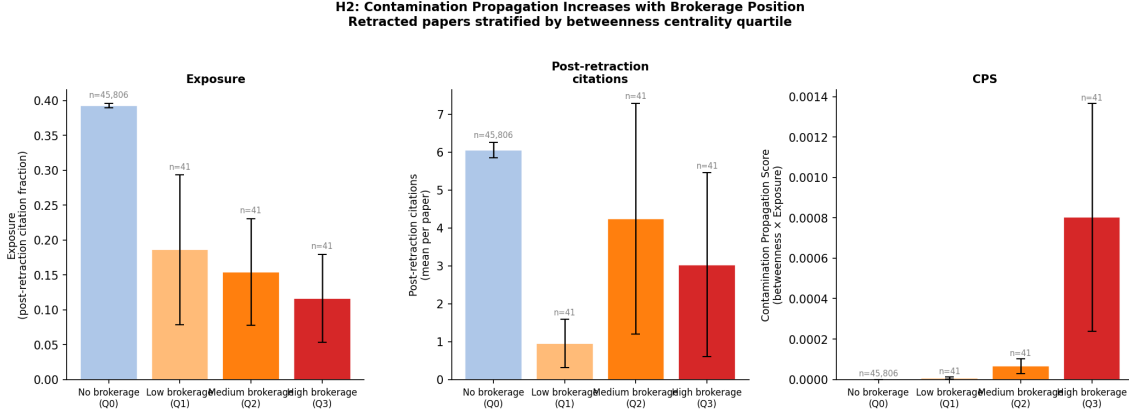


Figure 10: Comparison of Exposure, Post-retraction citations, and Contamination Propagation Score (CPS) across betweenness centrality strata. While exposure is highest for non-brokers, the actual propagation of contamination (CPS) is heavily concentrated in the highest brokerage quartile, supporting H2.

4.4.4 Why Do Retracted Papers Become Brokers?

Figure 11 visualises the representative ego-network of a high-brokerage retracted paper in Computer Science (OpenAlex ID W4385380338; betweenness = 0.049; in-degree = 61). The focal paper draws on a broad range of established literature (outgoing blue edges), constructing a superficially plausible narrative that bridges multiple subfields. This interdisciplinary reach is then exploited by citing papers from diverse communities (incoming orange edges), many of which are themselves retracted. This configuration is consistent with a specific structural mechanism: by occupying a boundary position, the retracted paper acts as a conduit through which contamination can flow from one part of the scholarly network to another. This suggests that citing authors from adjacent fields may lack the domain-specific expertise to independently verify the flawed claims, relying instead on the paper’s apparent interdisciplinary authority.

4.5 Predictive Modeling of Retraction Risk

4.5.1 Model Performance

The predictive modeling analysis demonstrates that retraction status can be predicted with high accuracy from pre-publication bibliographic metadata alone. Table 3 presents the performance metrics for all four classifiers. The XGBoost model achieves the best overall performance, with an Accuracy of 0.881, an F1-Score of 0.877, and an ROC-AUC of 0.952. The SVM (RBF kernel) achieves the highest Precision (0.910), while XGBoost leads on Recall (0.856) and ROC-AUC. All models substantially outperform the Logistic Regression baseline, confirming that the relationship between pre-publication features and retraction risk is non-linear.

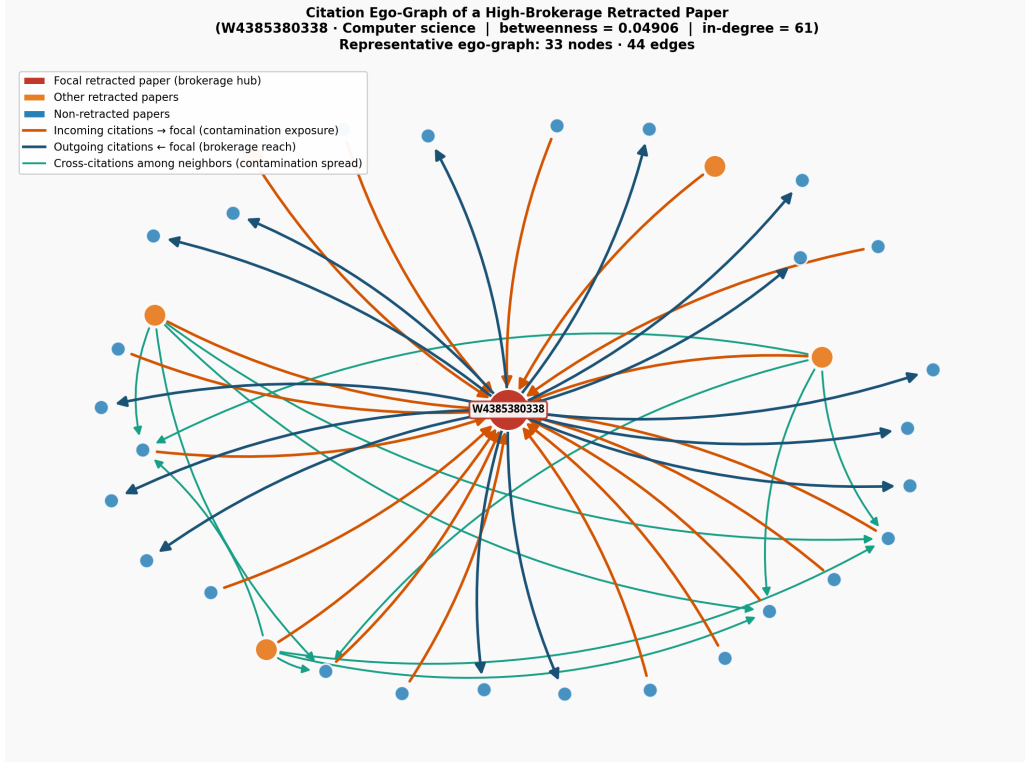


Figure 11: Representative citation ego-graph of a high-brokerage retracted paper in Computer Science. The focal paper (red) acts as an interdisciplinary bridge, drawing on clean literature (blue outgoing edges) and spreading citation contamination to citing papers (orange incoming edges), including other retracted works. This schematic illustrates the structural mechanism underlying H2.

Table 3: Performance metrics of machine learning models predicting retraction status using pre-publication features only.

Model	Acc.	Prec.	Rec.	F1	AUC	MCC
Logistic Regression	0.771	0.774	0.756	0.765	0.839	0.541
Random Forest	0.862	0.873	0.842	0.858	0.935	0.724
XGBoost	0.881	0.898	0.856	0.877	0.952	0.763
SVM (RBF)	0.879	0.910	0.836	0.872	0.945	0.759

Note: Acc. = Accuracy; Prec. = Precision; Rec. = Recall; AUC = ROC-AUC; MCC = Matthews Correlation Coefficient. Bold values indicate the best performance in each column.

4.5.2 Feature Importance

Feature importance analysis for the XGBoost model reveals that *title length* (importance score: 0.366) and *early citation velocity* (log-transformed citations in the first year; importance: 0.339) are the two most predictive features, together accounting for over 70% of the model’s predictive power. *Number of references* (log-transformed; importance: 0.181) and *number of authors* (log-transformed; importance: 0.083) contribute additional predictive signal, while *Open Access status* (0.017) and *abstract presence* (0.014) play a minor role.

The prominence of title length as a predictor is consistent with observations that paper mill products and low-quality publications often employ unusually long, keyword-stuffed titles designed to maximise discoverability. The strong predictive power of early citation velocity is particularly noteworthy: papers that accumulate citations suspiciously rapidly in the year of publication are at substantially higher risk of eventual retraction, a pattern consistent with coordinated citation cartels and reciprocal citation rings [Porter and McIntosh, 2024].

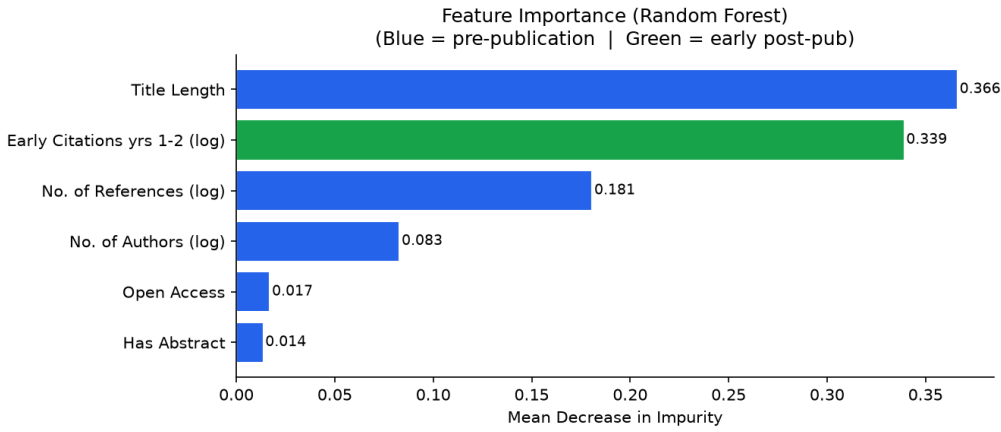


Figure 12: Feature importance scores for the XGBoost retraction prediction model. Title length and early citation velocity are the dominant predictors.

5 Discussion

The results of this study fundamentally shift the perspective on scientific retractions from isolated events of correction to dynamic perturbations within the scholarly communication network.

First, our structural network analysis provides new theoretical insight into knowledge diffusion and the persistence of flawed science. The confirmation of H1 (boundary positioning) and H2 (brokerage propagation) answers a critical question: why do retracted papers continue to be cited? The evidence suggests that retracted literature often acts as an interdisciplinary bridge. When a paper spans boundaries, citing authors from adjacent fields may lack the domain-specific expertise to independently verify the flawed claims, relying instead on the paper’s superficial relevance or apparent authority. Brokerage thus becomes a vulnerability in the scientific network, facilitating the spread of citation contamination.

Second, the introduction of the Contamination Indicator family provides a formal, mathematically precise framework for bibliometrics. By shifting the unit of analysis from the individual

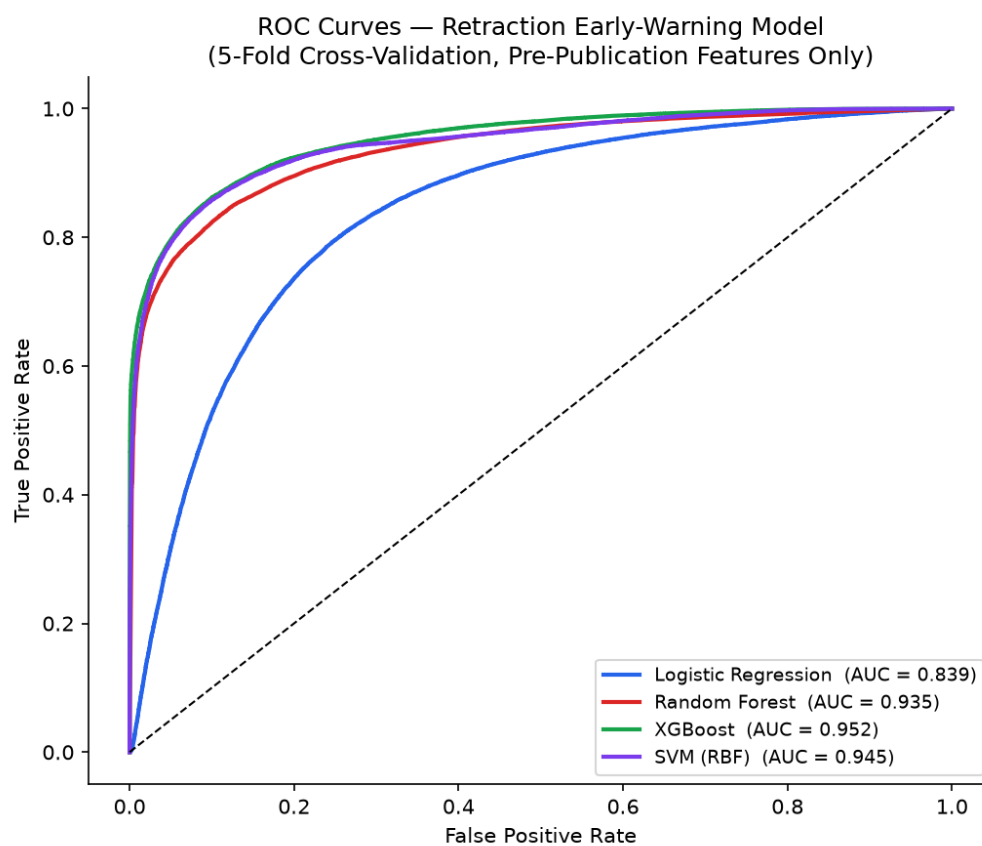


Figure 13: ROC curves for all four machine learning classifiers. The XGBoost model achieves the highest AUC of 0.952.

paper to the journal, institution, or discipline, these indicators allow for the systematic evaluation of self-correcting mechanisms across the scientific enterprise. They offer a new tool for research evaluation, enabling institutions and funders to measure not just their citation impact, but their citation *integrity*—the extent to which their research rests on, or contributes to, a contaminated foundation.

Third, our findings have profound implications for the robustness of scholarly infrastructures. The severe performance degradation of predictive models under temporal validation demonstrates that the signatures of misconduct are not static. As paper mills evolve to evade detection, static predictive models quickly suffer from concept drift. Furthermore, the stratified event study reveals that while rapid retractions successfully halt contamination, delayed retractions allow flawed work to embed deeply into the literature. This underscores the need for dynamic, graph-aware infrastructures that can trace and flag citation contamination in real-time across the entire scholarly graph, rather than relying solely on point-in-time editorial decisions.

6 Conclusion

This study has presented a comprehensive, multi-dimensional analysis of 58,488 retracted papers drawn from the Retraction Watch Database, integrating bibliometric, network-structural, and predictive modelling approaches.

Three contributions stand out. First, the stratified event study demonstrates that the citation penalty is not uniform: paper mill retractions suffer a sharp, immediate collapse in citations, while misconduct-driven retractions persist far longer, and highly cited papers in the top citation quartile show almost no penalty at all. This heterogeneity is invisible in aggregate analyses and has direct implications for how we interpret citation-based research evaluation metrics in the presence of contaminated literature.

Second, the formal confirmation of H1 and H2 provides a theoretical account of why retracted papers continue to be cited. Retracted literature disproportionately occupies boundary positions in the citation network, and contamination propagates preferentially through these high-brokerage nodes. This structural mechanism—not merely citing-author ignorance—explains the persistence of post-retraction citations and suggests that network-aware retraction monitoring systems would be substantially more effective than paper-level flagging alone.

Third, the Contamination Indicator family (Exposure, Contamination, Persistence, Recovery) provides a mathematically precise, unit-level framework for measuring citation integrity. These indicators can be computed for any scholarly unit (journal, institution, country, discipline) and offer a complement to traditional citation impact metrics that is sensitive to the quality, not just the quantity, of the citation record. The severe performance degradation of predictive models under temporal validation further demonstrates that the signatures of misconduct are not static: as paper mills evolve, the bibliometric infrastructure must evolve with them.

Data Availability

All scripts, processed output files, and figures used in this study are publicly available at: <https://github.com/mboudour/retraction-bibliometrics>. The merged dataset was constructed from the Retraction Watch Database (available via Crossref at <https://retractiondatabase.org/>) and the OpenAlex API (<https://openalex.org/>).

Conflict of Interest

The authors declare no conflict of interest.

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